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Question # 1 of 30

SOLVED BY: WAQAR HUSSAIN

If a particle is projected making a certain angle with horizontal into the air with a certain velocity or dropped from a moving plane then it is called a

Select the correct option

- | | | |
|----------------------------------|---------------|---|
| ☐ | hyperbola |  |
| ☒ | projectile |  |
| ☐ | circle |  |
| ☐ | Straight line |  |

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Question # 2 of 30

SOLVED BY: WAQAR HUSSAIN

The path of a projectile is called its

Select the correct option



velocity



trajectory



acceleration



none of these



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Question # 3 of 30

SOLVED BY: WAQAR HUSSAIN

Angle with which the particle is projected is called the _____ of projectile.

Select the correct option

- | | | |
|----------------------------------|---------------|---|
| ☐ | velocity |  |
| ☒ | angle |  |
| ☐ | acceleration |  |
| ☐ | none of these |  |

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Question # 4 of 30

SOLVED BY: WAQAR HUSSAIN

The velocity with which the particle is projected is called the _____ of projection.

Select the correct option

- | | | |
|----------------------------------|---------------|---|
| ☐ | acceleration |  |
| ☐ | angle |  |
| ☐ | none of these |  |
| ☒ | velocity |  |

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Question # 5 of 30

SOLVED BY: WAQAR HUSSAIN

If we neglect the air resistance, a projectile is a particle moving under the influence of _____ force.

Select the correct option

- | | | |
|----------------------------------|---------------|---|
| ☐ | frictional |  |
| ☐ | angle |  |
| ☐ | none of these |  |
| ☒ | gravitational |  |

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Question # 6 of 30

SOLVED BY: WAQAR HUSSAIN

If 45° and v_0 are the angle and velocity of projection respectively, then the equation of path of projectile is $y =$

Select the correct option

- | | | |
|----------------------------------|---|---|
| ☐ | $\sqrt{3}x - \frac{2gx^2}{v_0^2}$ |  |
| ☒ | $x - \frac{gx^2}{v_0^2}$ |  |
| ☐ | $\frac{x}{\sqrt{3}} - \frac{2gx^2}{3v_0^2}$ |  |
| ☐ | $x - gv_0^2x^2$ |  |

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Question # 7 of 30

SOLVED BY: WAQAR HUSSAIN

If 60° and v_0 are the angle and velocity of projection respectively, then the equation of path of projectile is $y =$

Select the correct option

- | | | |
|----------------------------------|---|---|
| ☐ | $x - \frac{gx^2}{v_0^2}$ |  |
| ☒ | $\sqrt{3}x - \frac{2gx^2}{v_0^2}$ |  |
| ☐ | $\frac{x}{\sqrt{3}} - \frac{2gx^2}{3v_0^2}$ |  |
| ☐ | $x - gv_0^2x^2$ |  |

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Question # 8 of 30

SOLVED BY: WAQAR HUSSAIN

If 30° and v_0 are the angle and velocity of projection respectively, then the equation of path of projectile is $y =$

Select the correct option

- | | | |
|----------------------------------|---|---|
| ☐ | $x - \frac{gx^2}{v_0^2}$ |  |
| ☒ | $\frac{x}{\sqrt{3}} - \frac{2gx^2}{3v_0^2}$ |  |
| ☐ | $\sqrt{3}x - \frac{2gx^2}{v_0^2}$ |  |
| ☐ | $x - gv_0^2x^2$ |  |

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Question # 9 of 30

SOLVED BY: WAQAR HUSSAIN

If 45° and $\frac{1}{3} m/s$ are the angle and velocity of projection respectively, then the equation of path of projectile is $y =$

Select the correct option

- | | | |
|----------------------------------|---|---|
| ☐ | $x - \frac{gx^2}{v_0^2}$ |  |
| ☐ | $\sqrt{3}x - \frac{2gx^2}{v_0^2}$ |  |
| ☒ | $x - 9gx^2$ |  |
| ☐ | $\frac{x}{\sqrt{3}} - \frac{2gx^2}{3v_0^2}$ |  |

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Question # 10 of 30

SOLVED BY: WAQAR HUSSAIN

The path of a projectile is a

Select the correct option

- | | | |
|----------------------------------|---------------|---|
| ☒ | parabola |  |
| ☐ | straight line |  |
| ☐ | circle |  |
| ☐ | none of these |  |

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Question # 11 of 30

SOLVED BY: WAQAR HUSSAIN

The maximum horizontal range of a projectile is

Select the correct option

- | | | |
|----------------------------------|----------------------------|---|
| ☐ | $\frac{v_0^2}{2g}$ |  |
| ☐ | $\frac{\sqrt{3}v_0^2}{2g}$ |  |
| ☐ | $v_0^2 g$ |  |
| ☒ | $\frac{v_0^2}{g}$ |  |

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Question # 13 of 30

SOLVED BY: WAQAR HUSSAIN

If 8 ft/s is the velocity of projection, then the maximum horizontal range is

Select the correct option

- | | | |
|----------------------------------|------|---|
| ☐ | 6 ft |  |
| ☒ | 2 ft |  |
| ☐ | 4 ft |  |
| ☐ | 3 ft |  |

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Question # 14 of 30

SOLVED BY: WAQAR HUSSAIN

If 80 ft/s is the velocity of projection, then the maximum horizontal range is

Select the correct option

- | | | |
|----------------------------------|--------|---|
| ☐ | 10 ft |  |
| ☒ | 200 ft |  |
| ☐ | 100 ft |  |
| ☐ | 250 ft |  |

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Question # 15 of 30

SOLVED BY: WAQAR HUSSAIN

The maximum horizontal range of a projectile is 100 ft, the velocity of projection is

Select the correct option

- | | | |
|----------------------------------|------------|---|
| ☐ | 3.6 ft/s |  |
| ☐ | 4.6 ft/s |  |
| ☐ | 5.6 ft/s |  |
| ☒ | 56.57 ft/s |  |

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Question # 16 of 30

SOLVED BY: WAQAR HUSSAIN

If 60° and v_o are the angle and velocity of projection respectively, then the time of flight of projectile is

Select the correct option

- | | | |
|----------------------------------|-------------------------|---|
| ☐ | $\frac{\sqrt{2}v_o}{g}$ |  |
| ☐ | $\frac{v_o}{g}$ |  |
| ☒ | $\frac{\sqrt{3}v_o}{g}$ |  |
| ☐ | $\frac{v_o}{\sqrt{2}g}$ |  |

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Question # 17 of 30

SOLVED BY: WAQAR HUSSAIN

If 45° and v_o are the angle and velocity of projection respectively, then the time taken to attain the maximum height of the projectile is

Select the correct option

- | | | |
|----------------------------------|-------------------------|---|
| ☒ | $\frac{v_o}{\sqrt{2}g}$ |  |
| ☐ | $\frac{\sqrt{2}v_o}{g}$ |  |
| ☐ | $\frac{v_o}{g}$ |  |
| ☐ | $\frac{\sqrt{3}v_o}{g}$ |  |

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Question # 18 of 30

SOLVED BY: WAQAR HUSSAIN

If 30° and v_o are the angle and velocity of projection respectively, then the time taken to attain the maximum height of the projectile is

Select the correct option

- | | | |
|----------------------------------|--------------------------|---|
| ☐ | $\frac{\sqrt{3}v_o}{2g}$ |  |
| ☐ | $\frac{v_o}{g}$ |  |
| ☒ | $\frac{v_o}{2g}$ |  |
| ☐ | none of these |  |

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Question # 19 of 30

SOLVED BY: WAQAR HUSSAIN

If 60° and v_o are the angle and velocity of projection respectively, then the time taken to attain the maximum height of the projectile is

Select the correct option

- | | | |
|----------------------------------|--------------------------|---|
| ☐ | $\frac{v_o}{2g}$ |  |
| ☐ | $\frac{v_o}{g}$ |  |
| ☐ | none of these |  |
| ☒ | $\frac{\sqrt{3}v_o}{2g}$ |  |

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Question # 20 of 30

SOLVED BY: WAQAR HUSSAIN

A projectile is projected with speed 30 m/s at an angle of 30° to the horizontal, the time of flight is

Select the correct option

☐	4 s	
☒	3 s	
☐	2 s	
☐	1 s	

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Question # 21 of 30

SOLVED BY: WAQAR HUSSAIN

Time of flight on an inclined plane of inclination $-\beta$ is $t =$

Select the correct option

- | | | |
|----------------------------------|--|---|
| ☒ | $\frac{2v_o \sin(\alpha + \beta)}{g \cos \beta}$ |  |
| ☐ | $\frac{2v_o \cos(\alpha + \beta)}{g \cos \beta}$ |  |
| ☐ | $\frac{2v_o \sin(\alpha - \beta)}{g \cos \beta}$ |  |
| ☐ | none of these |  |

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Question # 22 of 30

SOLVED BY: WAQAR HUSSAIN

The locus of all points on different trajectories which lie at maximum distances from point of projection is called _____ of safety.

Select the correct option

- | | | |
|----------------------------------|-----------|---|
| ☐ | hyperbola |  |
| ☒ | parabola |  |
| ☐ | ellipse |  |
| ☐ | circle |  |

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Question # 23 of 30

SOLVED BY: WAQAR HUSSAIN

The equation of parabola of safety is $x^2 =$

Select the correct option

- | | | |
|----------------------------------|---|---|
| ☐ | $-\frac{2v_o^2}{g} \left(y - \frac{2v_o^2}{g} \right)$ |  |
| ☐ | $-\frac{2v_o^2}{g} \left(y - \frac{v_o^2}{g} \right)$ |  |
| ☒ | $-\frac{2v_o^2}{g} \left(y - \frac{v_o^2}{2g} \right)$ |  |
| ☐ | none of these |  |

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SOLVED BY: WAQAR HUSSAIN

The directrix of the parabola of safety is

Select the correct option

- | | | |
|----------------------------------|--------------------|---|
| ☒ | $\frac{v_o^2}{g}$ |  |
| ☐ | $\frac{v_o^2}{4g}$ |  |
| ☐ | $\frac{v_o^2}{2g}$ |  |
| ☐ | $\frac{2v_o^2}{g}$ |  |

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SOLVED BY: WAQAR HUSSAIN

The vertex of the parabola of safety is

Select the correct option

- | | | |
|----------------------------------|------------------------------------|---|
| ☐ | $\left(0, \frac{4v_o^2}{g}\right)$ |  |
| ☐ | $\left(0, \frac{2v_o^2}{g}\right)$ |  |
| ☐ | $\left(0, \frac{v_o^2}{g}\right)$ |  |
| ☒ | $\left(0, \frac{v_o^2}{2g}\right)$ |  |

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Question # 26 of 30

SOLVED BY: WAQAR HUSSAIN

Areal speed of the radius vector $\vec{r}$ joining the particle to the centre of force is

Select the correct option



constant



not constant



variable



none of these



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Question # 27 of 30

SOLVED BY: WAQAR HUSSAIN

The component of force along the radius vector is

Select the correct option



$$F_{\theta} = \frac{m}{r} \frac{d}{dt} \left(r^2 \frac{d\theta}{dt} \right)$$



$$F_r = m \left[\frac{dr}{dt} - r \left(\frac{d\theta}{dt} \right) \right]$$



$$F_r = m \left[\frac{d^2 r}{dt^2} - r \left(\frac{d\theta}{dt} \right)^2 \right]$$



none of these



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Question # 28 of 30

SOLVED BY: WAQAR HUSSAIN

The component of force perpendicular to the radius vector is

Select the correct option



$$F_{\theta} = \frac{m}{r} \frac{d}{dt} \left(r^2 \frac{d\theta}{dt} \right)$$



$$F_r = m \left[\frac{d^2 r}{dt^2} - r \left(\frac{d\theta}{dt} \right)^2 \right]$$



$$F_r = m \left[\frac{dr}{dt} - r \left(\frac{d\theta}{dt} \right) \right]$$



none of these



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Question # 29 of 30

SOLVED BY: WAQAR HUSSAIN

When the force $\vec{F}$ is an attractive force directed towards origin, then $F_r = -F$ and $F_\theta =$

Select the correct option



0



3



1



2



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Question # 30 of 30

SOLVED BY: WAQAR HUSSAIN

The differential equation of the orbit in polar coordinates is $\frac{d^2u}{d\theta^2} + u =$

Select the correct option

- | | | |
|----------------------------------|--------------------|---|
| ☐ | $\frac{f}{hu}$ |  |
| ☒ | $\frac{f}{h^2u^2}$ |  |
| ☐ | $\frac{f}{h^3u^3}$ |  |
| ☐ | none of these |  |

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SOLVED BY: WAQAR HUSSAIN

The component of force along the radius vector is

Select the correct option

- | | | |
|-----------------------|---|---|
| ☐ | $F_{\theta} = \frac{m}{r} \frac{d}{dt} \left(r^2 \frac{d\theta}{dt} \right)$ |  |
| ☐ | $F_r = m \left[\frac{dr}{dt} - r \left(\frac{d\theta}{dt} \right) \right]$ |  |
| ☐ | $F_r = m \left[\frac{d^2 r}{dt^2} - r \left(\frac{d\theta}{dt} \right)^2 \right]$ |  |



Replies



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ahmad zia • 6 mo ago



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